

UniCAR-RL: Seeing Better before Thinking Deeper in Visual Mathematics

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<https://github.com/yzli9/UniCAR-RL>

Abstract

*Multimodal Large Language Models (MLLMs) often struggle with complex mathematical visual reasoning primarily due to a lack of fine-grained perception, causing initial visual hallucinations to directly trigger cascading reasoning failures. In traditional end-to-end reinforcement learning (RL), sparse rewards fail to decouple perceptual hallucinations from logical missteps, hindering targeted perception optimization. Alternatively, fine-tuning with perception-enhanced CoT data incurs high costs and hallucinations. In this paper, we address these challenges by proposing **UniCAR-RL**, an RL framework for annotation-free perception optimization. By explicitly decoupling the optimization of perception and reasoning during the training process, it achieves isolation and optimization of both capabilities. Specifically, UniCAR-RL consists of three synergistic branches: 1) a Caption-RL branch that optimizes perception capabilities through verifier-guided reasoning validation; 2) a Reasoning-RL branch that performs logical reasoning based on a gold image description to halt cascading errors; 3) a QA-RL branch that retains native end-to-end alignment to ensure robust question-answering performance. Experiments show that UniCAR-RL substantially improves MLLMs' mathematical and visual reasoning using only raw short-answer data. Furthermore, it demonstrates strong generalization across diverse architectures and scales.*

1. Introduction

Driven by the rapid advancements in Chain-of-Thought (CoT) [1, 2] and Reinforcement Learning (RL) [3, 4] techniques, Multimodal Large Language Models (MLLMs) have achieved substantial progress in their reasoning capabilities for complex tasks [5–7]. However, the performance of

existing MLLMs remains unsatisfactory when confronted with abstract visual reasoning tasks, such as math [8, 9] and logic [10, 11]. Extensive research [12–14] shows that the primary bottleneck is not the models' insufficient reasoning skills, but rather the absence of a fine-grained perception capability. As illustrated in Fig. 1(a), when processing complex mathematical figures, models may easily misinterpret critical yet microscopic visual conditions, causing subsequent rigorous mathematical reasoning to be founded upon hallucinated visual premises. These initial visual misinterpretations fundamentally cause the breakdown of the mathematical reasoning.

As illustrated in Fig. 1(b), extensive recent research has emphasized the importance of enhancing fine-grained visual perception capabilities in MLLMs. Some recent works [15, 16] attempt to optimize models' perceptual abilities by distilling high-quality, perception-enhanced CoT data. However, these methods heavily rely on costly expert-level annotations and are inherently constrained by the hallucinations of teacher models. Alternatively, as an effective paradigm for boosting model performance on complex tasks, RL has been employed for optimization. However, sparse global rewards in RL fail to distinguish whether an incorrect prediction stems from visual misperception or logical errors, thereby rendering perception optimization highly inefficient.

To address the above challenges of existing methods, we propose Unified Caption-Answer-Reasoning Reinforcement Learning (**UniCAR-RL**), an RL framework for annotation-free perception optimization. As illustrated in Fig. 1(b), UniCAR-RL fundamentally disentangles perceptual evaluation from end-to-end black-box training by introducing an explicit image description process and an advanced verifier.

Specifically, to obtain a pure perceptual feedback signal, the **Caption-RL Branch** of UniCAR-RL isolates the MLLM's internal reasoning module to generate an exclusive image description, and directly utilizes an advanced verifier's evaluation of this description as the reward. If the verifier successfully deduces the ground-truth answer relying solely on image description, it confirms the flawless extrac-

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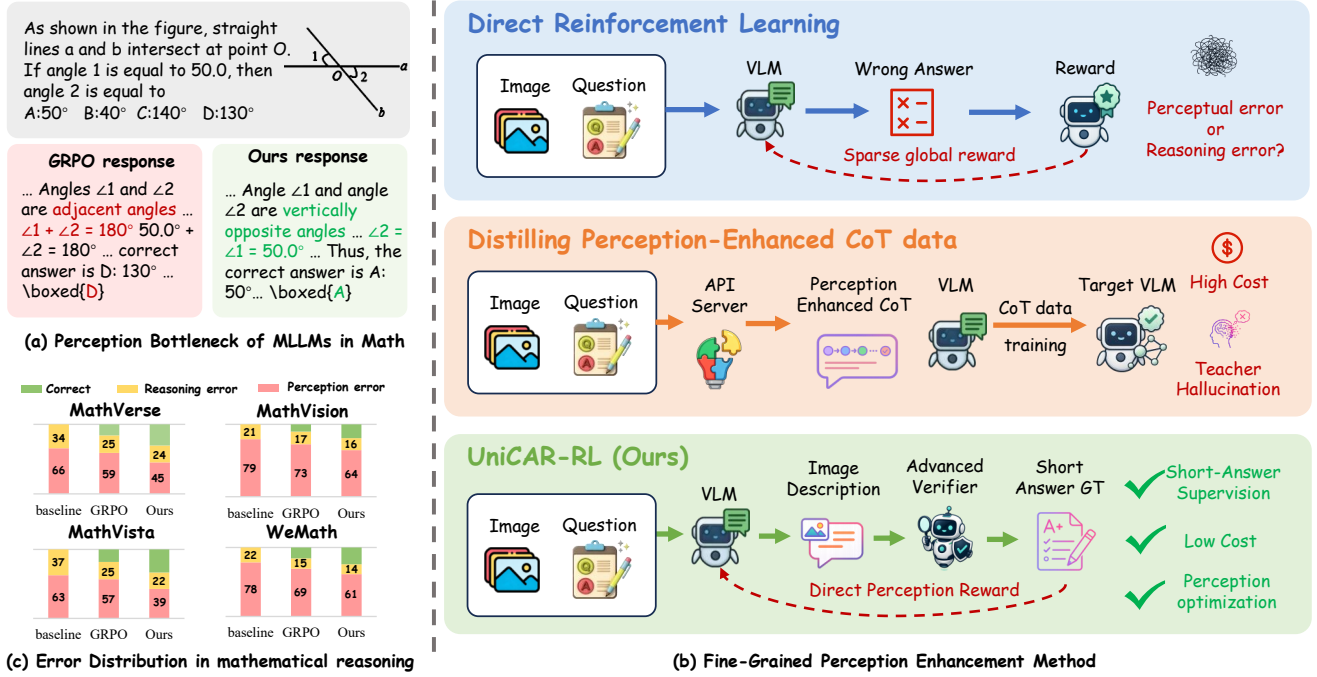


Figure 1. Motivation of proposed approach. (a) highlights the perception bottleneck of MLLMs in visual mathematical reasoning, but existing RL methods struggle to address it; (b) presents existing perception enhancement methods and their limitations; (c) provides a comparison of error distributions on visual mathematical reasoning tasks.

tion of visual information. Consequently, this verification mechanism returns an unambiguous direct perception reward to MLLMs, enabling precise optimization of perception capabilities while eliminating the interference of reasoning errors.

Building upon the high-quality image descriptions, **Reasoning-RL Branch** further enhances the model’s logical reasoning capabilities. At this stage, this branch forces the model to perform independent mathematical reasoning based strictly on the gold caption verified in the preceding stage. This design of visual isolation fundamentally severs the error cascade of fine-grained perceptual hallucinations into the reasoning process, ensuring optimization signals are purely and intensely focused on enhancing logical deduction capabilities.

Finally, **QA-RL Branch** jointly optimizes perception and reasoning end-to-end, alongside maintaining the model’s inherent question-answering proficiency. By synergizing these three branches, UniCAR-RL successfully achieves the joint optimization of perception and reasoning based on original short answers. The main advantages of UniCAR-RL are summarized as follows:

1) **Perception Enhancement via Raw Short-Answer Data.** As shown in Fig. 1(c), unlike traditional RL that primarily optimizes reasoning, this framework relies solely

on raw short answers to simultaneously improve perceptual capabilities while preserving reasoning optimization.

2) **Performance Gains on Mathematical Tasks.** This framework achieves comprehensive gains across diverse mathematical benchmarks, delivering performance comparable to close-source proprietary models at a fraction of the training cost.

3) **Generalization Across Different Architectures and Sizes.** This framework achieves consistent and significant performance improvements across varying architectures and sizes. Furthermore, optimization efficacy exhibits limited sensitivity to the capabilities of the advanced verifier.

2. Related Work

RL for Mathematical Reasoning. As reinforcement learning continually advances mathematical reasoning, GRPO [3] and its variants [17–19] have become widely used post-training paradigms in this domain. In multimodal mathematical reasoning, existing methods fall into cold-start RL and zero-RL. Cold-start RL bootstraps policies via curated reasoning traces or process supervision. We-Math 2.0 [20] structures training around mathematical knowledge, Vision-R1 [21] builds multimodal CoT data for GRPO initialization, and URSA [22] combines a multimodal CoT foundation with process-supervised GRPO. Zero-RL directly opti-

mizes MLLMs with rule-based rewards. MM-Eureka [23] scales rule-based RL on filtered image-text math data, while Shuffle-R1 [4] improves RL efficiency through data-centric trajectory sampling and batch shuffling. R1-VL [24] introduces StepGRPO with dense step-wise rewards, and VL-Cogito [25] adopts curriculum RL with dynamic length rewards.

Visual Perception Enhancement in MLLMs. Fine-grained visual perception is essential for complex visual tasks, yet remains a major bottleneck for MLLMs [13, 16, 26]. Existing methods mainly improve visual perception through perception-augmented trajectory training and perception-rewarded RL. The former typically distills high-quality perception-enhanced CoT data from advanced teacher models and then uses these data to enhance the perceptual capabilities of MLLMs. CodePercept [15] and GeoCode [27] use executable programs to recover diagram structures, while MathFlow [26] decouples visual extraction from downstream inference. ViRC [28] and Geoint-R1 [29] further structure the reasoning process with grounded reasoning chunks and auxiliary geometric constructions. Perception-reward-guided RL methods [30–32] incorporate perceptual signals into reward design. They introduce rewards based on visual annotation consistency, visual knowledge internalization, and caption utility, respectively. Other studies reduce reliance on distilled perception data: VPPO [33] focuses policy updates on visually dependent tokens, Vision-SR1 [14] self-verifies generated visual descriptions, PAPO [34] derives grounding signals from visual perturbations, and NoisyRollout [35] leverages noisy visual rollouts. Details of those works are provided in Appendix C.

3. Method

3.1. Overall Framework

Visual mathematical reasoning demands two tightly coupled competencies: *perception* faithfully extracting fine-grained visual details such as axis values, geometric marks, and symbolic annotations, and *reasoning* chains logical deductions toward a correct answer. A model can fail for either reason, yet conventional outcome-driven RL treats both failure modes identically: a wrong final answer yields a negative signal regardless of its root cause. Without a targeted signal, the perception bottleneck silently persists across training.

The key insight that motivates our approach is intuitive: *the quality of a visual description can be objectively measured by whether a capable text-only solver can derive the correct answer from it alone.* This reformulation converts the inherently subjective problem of evaluating perception into a clean, annotation-free binary signal tied directly to task utility rather than surface-level textual fidelity.

Building on this insight, we propose **UniCAR-RL**, disentangling the optimization into three jointly updated branches. As illustrated in Fig. 2, the **Caption-RL Branch** encourages broader, question-agnostic visual perception and evaluates generated descriptions by their task sufficiency. The **Reasoning-RL Branch** isolates logical reasoning in a purely textual setting, using the best available description as input. The **QA-RL Branch** then anchors both capabilities back to the native end-to-end task format. All three branches share the same policy parameters θ and are updated jointly in every iteration, so perception and reasoning co-evolve rather than being trained in isolated stages.

3.2. Caption-RL Branch

Motivation. The Caption-RL Branch is built on a deliberate act of information withholding: the question q is explicitly excluded from the model’s input. A model that knows the question can selectively attend only to visually relevant elements; a model that does not must describe everything. By stripping q from the context, we exert a natural pressure toward holistic, unbiased visual grounding.

Stage 1: Caption Generation. The policy π_θ generates G caption rollouts from the image I and a task prompt p_{cap} , tasked with exhaustively extracting visual details without producing an answer:

$$c_i \sim \pi_\theta(\cdot \mid I, p_{\text{cap}}), \quad i = 1, \dots, G. \quad (1)$$

This design is inspired by the intuition behind EM-style optimization: the caption acts as a latent intermediate explanation, which is first inferred under the current policy and then used to optimize downstream reasoning. We do not perform EM in a strict probabilistic sense, since c^* is chosen by reward-guided selection rather than posterior expectation, and the overall objective remains RL-based.

Stage 2: Verification. Each caption c_i is evaluated by a *proxy solver* $\mathcal{V}$, implemented as a text-only language model that accepts no visual input by design. Given only the caption c_i , the question q , and a task prompt p_{ver} , $\mathcal{V}$ attempts to solve the problem entirely from the textual evidence at hand:

$$\hat{a}_i = \mathcal{V}(c_i, q, p_{\text{ver}}), \quad i = 1, \dots, G. \quad (2)$$

Because $\mathcal{V}$ is a capable text-based reasoner with no access to I , the resulting reward serves as a practical proxy for caption sufficiency, though it may still contain noise from the verifier itself. The perception reward thus becomes:

$$r_i^{\text{cap}} = \mathbb{I}[\mathcal{D}(\text{Extract}(\hat{a}_i), a^*)], \quad (3)$$

where a^* denotes the ground-truth answer. Missing a single critical visual element, such as a tick label, a right-angle mark, or a legend entry, will cause the text-only proxy solver to fail, sending a precise corrective signal back to the policy.

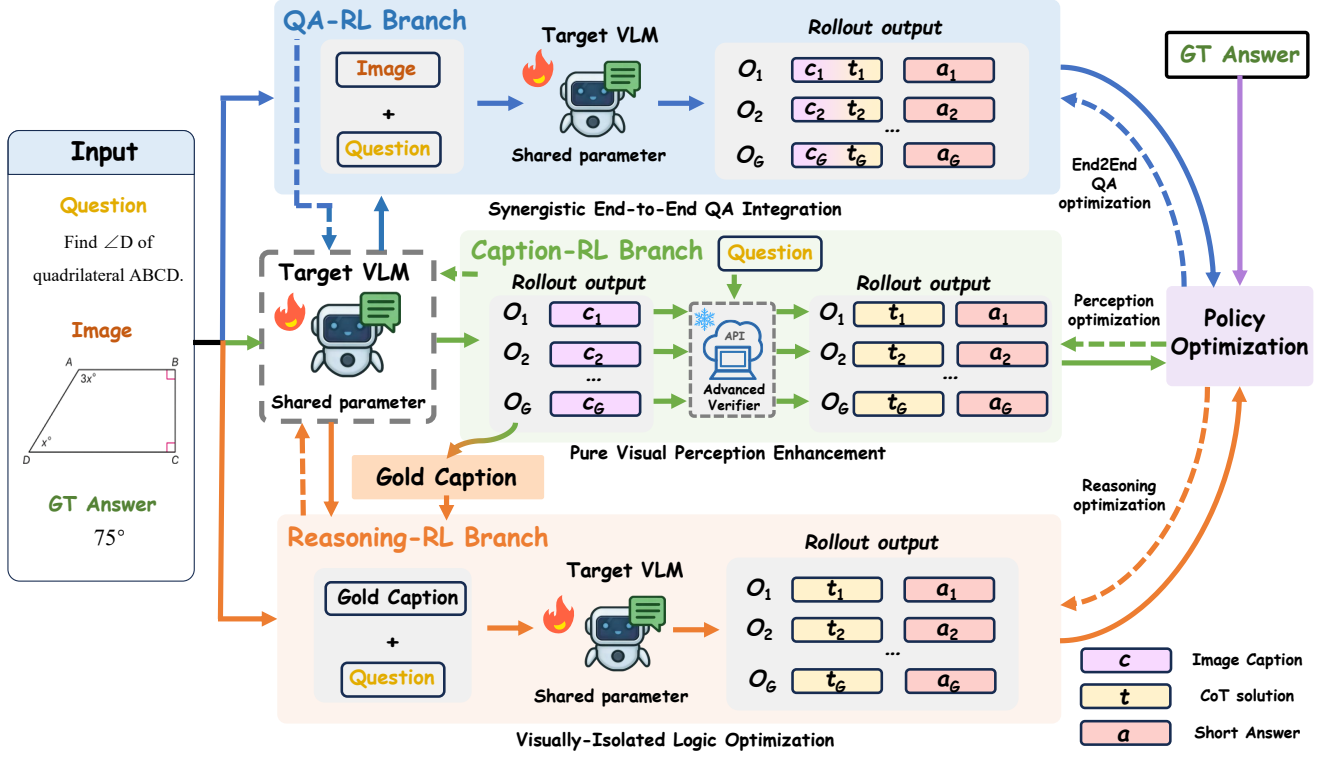


Figure 2. Overview of UniCAR-RL. It disentangles optimization into three branches: a) the **Caption-RL Branch** for pure visual perception enhancement, b) the **Reasoning-RL Branch** for visually-isolated logic optimization, and c) the **QA-RL Branch** for synergistic end-to-end QA integration.

3.3. Reasoning-RL Branch

Gold Caption Selection. From the G caption rollouts, we select the highest-reward caption as the selected *gold caption* c^* :

$$c^* = \arg \max_{c_i} r_i^{\text{cap}}, \quad (4)$$

breaking ties uniformly at random. Conditioning on c^* is deliberate: it provides a description that is simultaneously realistic (drawn from the policy’s own distribution) and maximally informative.

Text-Only Reasoning. The image is withheld, and the model solves the problem from c^* , q and a task prompt p_{rsn} :

$$a'_j \sim \pi_{\theta}(\cdot \mid c^*, q, p_{\text{rsn}}), \quad j = 1, \dots, G. \quad (5)$$

This setup is structurally symmetric to the verification step in Caption-RL: there, the proxy solver $\mathcal{V}$ takes a caption and derives an answer; here, π_{θ} does the same but as the learner, so gradients flow directly through this rollout. In effect, the model is learning to be its own proxy solver. The reward

follows the same rule-based extraction:

$$r_j^{\text{rsn}} = \mathbb{I}[\mathcal{D}(\text{Extract}(a'_j), a^*)]. \quad (6)$$

This branch also induces a *natural curriculum*: as **Caption-RL Branch** improves, c^* becomes progressively richer, presenting **Reasoning-RL Branch** with increasingly demanding textual inputs. Stronger perception raises the ceiling for reasoning; stronger reasoning amplifies the value of accurate perception.

3.4. QA-RL Branch

The two specialized branches target perception and reasoning separately. The **QA-RL Branch** serves as an integration stage: it trains the model to deploy both competencies under conditions matching downstream inference. The input image I , question q , and prompt p_{qa} mirror the test-time setting:

$$a_i \sim \pi_{\theta}(\cdot \mid I, q, p_{\text{qa}}), \quad i = 1, \dots, G, \quad (7)$$

$$r_i^{\text{qa}} = \mathbb{I}[\mathcal{D}(\text{Extract}(a_i), a^*)]. \quad (8)$$

By running this branch alongside the two specialized branches in every training iteration, the policy is continuously pulled back toward the task distribution it faces at

deployment. The decoupled branches sharpen individual competencies; the QA branch ensures those competencies are always being woven back together.

3.5. Unified GRPO Optimization

The three branches yield reward streams with heterogeneous semantics. To prevent any single branch from dominating the gradient, advantages are normalized independently within each rollout group. For branch $k \in \{\text{cap}, \text{rsn}, \text{qa}\}$:

$$A_i^{(k)} = \frac{r_i^{(k)} - \text{mean}(\mathbf{r}^{(k)})}{\text{std}(\mathbf{r}^{(k)})}. \quad (9)$$

Each branch’s objective takes the standard clipped GRPO:

$$J_k(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i^{(k)}|} \sum_{t=1}^{|o_i^{(k)}|} \min \left(r_{i,t}^{(k)} A_i^{(k)}, \right. \right. \\ \left. \left. \text{clip}(r_{i,t}^{(k)}, 1-\epsilon, 1+\epsilon) A_i^{(k)} \right) \right], \quad (10)$$

where the token-level importance ratio is

$$r_{i,t}^{(k)} = \frac{\pi_\theta(o_{i,t}^{(k)} \mid x^{(k)}, o_{i,<t}^{(k)})}{\pi_{\text{old}}(o_{i,t}^{(k)} \mid x^{(k)}, o_{i,<t}^{(k)})}. \quad (11)$$

All three objectives are optimized jointly via a single backward pass:

$$\theta \leftarrow \theta + \eta \nabla_\theta (J_{\text{cap}}(\theta) + J_{\text{rsn}}(\theta) + J_{\text{qa}}(\theta)). \quad (12)$$

This *locally decoupled, globally co-evolving* paradigm is the defining characteristic of UniCAR-RL: each branch contributes a specialized gradient signal targeting a distinct competency, yet all signals are applied simultaneously to the same shared parameters within every single update step.

4. Experiments

4.1. Experimental Setup

Dataset. We utilize 30k samples from MMRL30k [4] during the training. In the evaluation phase, we conduct extensive testing on four highly challenging multimodal mathematical reasoning benchmarks, including WeMath [42], MathVerse [8], MathVista [43], and MathVision [9]. Furthermore, we verify the model’s general capabilities on ChartQA [44] and HallusionBench [45].

Implementation Details. Our experiments use Qwen2.5-VL [39] and Qwen3-VL [38] as backbones for full-parameter fine-tuning. Models are optimized using AdamW [46] with a learning rate of 1×10^{-6} , a weight decay of 1×10^{-2} , a micro-batch size of 8 per device, and a rollout number of 16

per input. The text-only training verifiers are Qwen3.5-35B-A3B [47] and Gemini 3 Pro. For all locally evaluated models and benchmarks, decoding uses temperature 0.0, top- p 0.8, top- k 20, and presence penalty 0.0. For fair comparison, all evaluated models receive only the image and question as input during inference. Gemini-2.5-Flash [48] is used only as a benchmark answer judge when required by the evaluation protocol.

4.2. Main Results

Table 1 compares the proposed model with prior methods across four mathematical visual reasoning benchmarks. We draw the following observations:

1) **Significant performance gains.** By alleviating inherent perception bottlenecks of MLLMs, UniCAR-RL consistently delivers significant improvements across all evaluated backbone and scale configurations on visual mathematical reasoning benchmarks. It further achieves state-of-the-art average performance among the compared open-source models at comparable scales and remains competitive with proprietary models.

2) **Generalization across diverse backbones.** By inherently enhancing visual perception through decoupling, UniCAR-RL demonstrates robust generalization, achieving substantial performance gains across architectures and scales. Notably, baselines with smaller sizes and lower initial performance tend to yield more significant improvements.

3) **Effect of advanced verifier.** Experiments clearly reveal the relationship between advanced verifier capability and model performance. A stronger verifier provides more precise perception reward signals, thereby more accurately guiding the model’s optimization and significantly elevating the upper bound of its reasoning performance.

4.3. Visual Perception Evaluation

To quantitatively verify the perceptual enhancements of UniCAR-RL, we conduct a visual perception evaluation experiment. Specifically, we utilize the model to generate image descriptions for mathematical images, which are subsequently fed into a unified advanced verifier for problem-solving. Under this setup, the verifier’s accuracy objectively reflects the perception capability of the model. Table 3 demonstrates that while standard GRPO brings negligible improvements, UniCAR-RL secures significant breakthroughs in description quality.

4.4. General Tasks Evaluation

Although primarily optimized on visual mathematical data, UniCAR-RL mitigates the perception bottlenecks of MLLMs, yielding fine-grained perceptual enhancements that broadly benefit general multimodal tasks. We therefore further evaluate its generalization on a broader set of

Model	Venue	WeMath	MathVerse	MathVista	MathVision	Avg.
<i>Close-source Proprietary model</i>						
GPT-4o [36]		68.8	50.2	63.8	30.4	53.4
Claude-3.7-Sonnet [37]		72.6	52.0	66.8	41.3	58.1
<i>Models based on Qwen3-VL-Instruct</i>						
Qwen3-VL-4B [38]	arXiv 2025	75.0	47.2	73.5	50.1	61.5
Qwen3-VL-8B [38]	arXiv 2025	<u>79.4</u>	<u>60.8</u>	77.9	52.9	<u>67.8</u>
UniCAR-RL-4B (Gemini 3 Pro)	Ours	78.5	51.2	80.4	<u>53.3</u>	65.9
UniCAR-RL-8B (Gemini 3 Pro)	Ours	82.5	63.4	<u>79.5</u>	54.6	70.0
<i>Models based on Qwen2.5-VL-Instruct-3B</i>						
Qwen2.5-VL-3B [39]	arXiv 2025	55.9	33.6	59.4	22.5	42.9
MMR1-3B-RL* [40]	CVPR-F 2026	64.5	35.3	65.3	23.5	47.2
PAPO-D-3B* [34]	ICLR 2026	65.3	42.0	64.8	26.4	49.6
Shuffle-R1-3B* [4]	ICLR 2026	64.8	40.5	67.7	26.3	49.9
UniCAR-RL-3B (Qwen3.5-35B-A3B)	Ours	<u>66.5</u>	<u>41.8</u>	<u>68.9</u>	<u>27.3</u>	<u>51.1</u>
UniCAR-RL-3B (Gemini 3 Pro)	Ours	67.3	42.9	70.1	28.5	52.2
<i>Models based on Qwen2.5-VL-Instruct-7B</i>						
Qwen2.5-VL-7B [39]	arXiv 2025	61.7	42.5	67.3	25.7	48.3
MMR1-7B-RL* [40]	CVPR-F 2026	71.5	46.0	70.8	29.3	54.4
MM-Eureka-7B [†] [23]	TMLR 2026	66.1	50.3	73.0	26.9	54.0
NoisyRollout-7B [†] [35]	NeurIPS 2025	70.8	50.1	70.9	28.0	54.9
R1-ShareVL-7B* [41]	NeurIPS 2025	69.8	52.8	75.1	29.5	56.8
PAPO-D-7B* [34]	ICLR 2026	69.3	49.3	74.5	29.6	55.7
Vision-R1-7B* [21]	ICLR 2026	-	50.4	71.2	-	-
Vision-SR1-7B* [14]	ICLR 2026	63.9	43.5	68.4	27.4	50.8
Shuffle-R1-7B* [4]	ICLR 2026	68.7	49.3	75.2	30.0	55.8
Perception-R1-7B [†] [30]	ICLR 2026	<u>72.0</u>	54.3	74.2	28.6	57.2
UniCAR-RL-7B (Qwen3.5-35B-A3B)	Ours	71.9	53.0	<u>75.5</u>	<u>30.2</u>	<u>57.7</u>
UniCAR-RL-7B (Gemini 3 Pro)	Ours	72.4	<u>53.7</u>	76.0	30.6	58.2

Table 1. Main results on visual mathematical reasoning benchmarks. The Avg. score is the arithmetic mean over WeMath, MathVerse, MathVista, and MathVision; CVPR-F denotes CVPR Findings. Models marked with * are evaluated under our setting. Results marked with [†] are sourced from previous work [30]. The best value in each column is shown in **bold**, and the second-best is underlined. For UniCAR-RL (Verifier), the Verifier denotes the advanced verifier used during training. This convention applies to all subsequent tables.

general tasks, using the system prompts provided in Appendix E. Table 4 reports the results on ChartQA [44] and HallusionBench [45], while additional results are provided in Appendix D. The results show consistent performance gains across different model scales, demonstrating that the fine-grained perception enhancements of UniCAR-RL transfer beyond visual mathematical reasoning to diverse visual tasks.

4.5. Ablation Study

To investigate the contributions of each branch and component, we perform comprehensive ablation studies with Qwen3-VL-4B as the default setup. **Ablation on Different Branches.** To verify the contribution of each branch, we progressively remove the **Reasoning-RL Branch**, **Caption-RL**

Branch, and **QA-RL Branch**, gradually degrading UniCAR-RL back to the baseline. As demonstrated in Table 5, the absence of any single branch leads to performance degradation, proving that all three branches play a crucial synergistic role in enhancing reasoning capabilities. More importantly, compared to the mere 1.3% gain yielded by vanilla GRPO, the decoupled optimization of perception and reasoning significantly enhances the model’s reasoning capabilities, achieving a 3.1% improvement.

Ablation on Key Components. We further investigate the impact of different baselines, model sizes and verifiers on UniCAR-RL. As shown in Table 2, the results reveal three key observations. 1) First, UniCAR-RL consistently yields significant performance enhancements across various base models. 2) Second, this positive trend remains highly stable

Configuration	WeMath	MathVerse	MathVista	MathVision	Avg.
<i>Base model ablations (Verifier: Qwen3.5-35B-A3B)</i>					
Qwen2.5-VL-3B (base)	55.9	33.6	59.4	22.5	42.9
+ UniCAR-RL	66.5 (+10.6)	41.8 (+8.2)	68.9 (+9.5)	27.3 (+4.8)	51.1 (+8.2)
Qwen3-VL-4B (base)	75.0	47.2	73.5	50.1	61.5
+ UniCAR-RL	77.8 (+2.8)	50.4 (+3.2)	79.6 (+6.1)	52.9 (+2.8)	65.2 (+3.7)
<i>Verifier ablations</i>					
Qwen3-VL-4B (base)	75.0	47.2	73.5	50.1	61.5
w/ Qwen3.5-35B-A3B	77.8 (+2.8)	50.4 (+3.2)	79.6 (+6.1)	52.9 (+2.8)	65.2 (+3.7)
w/ Gemini 3 Pro	78.5 (+3.5)	51.2 (+4.0)	80.4 (+6.9)	53.3 (+3.2)	65.9 (+4.4)
<i>Parameter scale ablations (Verifier: Qwen3.5-35B-A3B)</i>					
Qwen3-VL-4B (base)	75.0	47.2	73.5	50.1	61.5
+ UniCAR-RL	77.8 (+2.8)	50.4 (+3.2)	79.6 (+6.1)	52.9 (+2.8)	65.2 (+3.7)
Qwen3-VL-8B (base)	79.4	60.8	77.9	52.9	67.8
+ UniCAR-RL	82.5 (+3.1)	63.4 (+2.6)	79.5 (+1.6)	54.6 (+1.7)	70.0 (+2.3)

Table 2. Component ablations of UniCAR-RL. We adopt Qwen3-VL-4B (Qwen3.5-35B-A3B) as the default configuration. We then conduct ablations on the base model, parameter scale, and verifier selection. Rows marked (base) denote the pretrained model without UniCAR-RL training and green values report absolute improvement over the corresponding base.

Captioner	Verifier-Solve Accuracy (%) $\uparrow$				
	WeMath	MathVerse	MathVista	MathVision	Avg.
Qwen3-VL-4B (zero-shot)	80.8	62.4	76.6	65.9	71.4
Qwen3-VL-4B + GRPO	82.1 (+1.3)	63.8 (+1.4)	77.8 (+1.2)	67.5 (+1.6)	72.8 (+1.4)
Qwen3-VL-4B + UniCAR-RL	84.9 (+4.1)	65.5 (+3.1)	82.5 (+5.9)	70.8 (+4.9)	75.9 (+4.5)

Table 3. Evaluation results of visual perception. We adopt Gemini 3 Pro as the unified verifier.

Model	ChartQA	HallBench
<i>Models around 4B parameters</i>		
Qwen3-VL-4B	83.0	73.7
+UniCAR-RL(Qwen3.5-35B-A3B)	87.9 (+4.9)	75.0 (+1.3)
+UniCAR-RL(Gemini 3 Pro)	89.5 (+6.5)	75.5 (+1.8)
<i>Models around 8B parameters</i>		
Qwen3-VL-8B	88.4	75.0
+UniCAR-RL(Qwen3.5-35B-A3B)	90.2 (+1.8)	75.2 (+0.2)
+UniCAR-RL(Gemini 3 Pro)	90.6 (+2.2)	75.4 (+0.4)

Table 4. Performance comparison on general multimodal benchmarks. HallBench stands for HallusionBench.

across different parameter scales, verifying the strong scalability of our approach. 3) Third, the verifier serves purely as an auxiliary for problem-solving. Although a stronger verifier inherently leads to better performance, the marginal gains are relatively limited, suggesting that powerful open-source models can effectively substitute for proprietary ones in most scenarios.

4.6. Qualitative Analysis

To illustrate the advantages of UniCAR-RL, we present representative comparative examples in Fig. 3. In mathematical reasoning tasks, the failures of the baseline model primarily stem from visual perception errors and logical reasoning errors. Although standard RL methods, represented by GRPO, can help correct reasoning errors, they still fall short in optimizing the perception capabilities of MLLMs. In contrast, benefiting from the explicit decoupling of the perception and reasoning processes, UniCAR-RL can simultaneously and specifically optimize both visual perception and logical reasoning capabilities. Its synergistic multi-branch architecture not only retains the error-correction advantages of RL, but also precisely rectifies visual misjudgments through an independent branch, ultimately achieving substantial improvements in overall performance.

4.7. Training Efficiency Analysis

To evaluate the training overhead associated with the performance gains of UniCAR-RL, we further analyze its training efficiency. We use Qwen2.5-VL-7B as the default backbone and measure both the rollout time and the total step time for

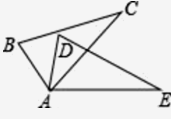
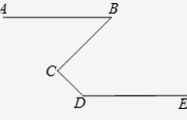
	Math problem	Baseline	GRPO	UniCAR-RL
Flawed Perception	If angle B = 70.0, angle C = 30.0, angle DAC = 35.0, then the degree of angle EAC is A:40° B:45° C:35° D:25°  Answer: B	... From the diagram, point D lies on line AE, so A, D, and E are collinear. ✗ False collinearity perception Therefore, ray AE coincides with ray AD, meaning the angle with AC at A is the same. $\angle EAC = \angle DAC$ So, $\angle EAC = 35^\circ$. Therefore, Final Answer: \boxed{C}	... the ray AD is the same direction as AE, meaning the angle formed with AC is identical whether we use AD or AE. So, $\angle EAC = \angle DAC$. ✗ False collinearity perception ... Substituting the given value: $\angle EAC = 35^\circ$. Therefore, the value of $\angle EAC$ is 35°. Final Answer: \boxed{C}	... The angles provided are: $\angle B = 70.0^\circ$ $\angle C = 30.0^\circ$ $\angle DAC = 35.0^\circ$... $\angle BAC = 180^\circ - 100^\circ = 80^\circ$... $\angle EAC = \angle BAC - \angle DAC = 80^\circ - 35^\circ = 45^\circ$... because $\angle EAC$ is the angle from EA to AC, while $\angle DAC$ is the angle from DA to AC ... Thus, the answer is B: 45°. Final Answer: \boxed{B}
	AB parallel DE, angle B = 30.0, angle C = 110.0, the degree of angle D is A:115° B:120° C:100° D:80°  Answer: C	... since CF is parallel to DE, and CD is a transversal, then the angle between CD and CF (which is 80°) is equal to the alternate interior angle with angle D. Therefore, angle D = 80°. ✗ Angle misclassification ... Answer: D Final Answer: \boxed{D}	We are given that AB is parallel to DE, $\angle B = 30^\circ$, and $\angle C = 110^\circ$... Introduce an auxiliary line. Through point C, draw a line parallel to AB and DE. Call this line CF, extending to the right ... $\angle BCF = \angle B = 30^\circ$... $\angle FCD = 80^\circ$... $\angle FCD + \angle D = 180^\circ$, $\angle D = 100^\circ$. Final Answer: \boxed{C}	We are given that AB is parallel to DE, and we know $\angle B = 30^\circ$, $\angle C = 110^\circ$... draw a line through point C that is parallel to both AB and DE. Let's call this line FG, passing through C ... $\angle B = \angle BCF = 30^\circ$... $\angle C = \angle BCF + \angle FCD \rightarrow 110^\circ = 30^\circ + \angle FCD \rightarrow \angle FCD = 110^\circ - 30^\circ = 80^\circ$... $\angle D + \angle FCD = 180^\circ \rightarrow \angle D = 100^\circ$ Final Answer: \boxed{C}

Figure 3. Qualitative comparison of mathematical reasoning tasks.

Configuration	WeMath	MathVerse	MathVista	MathVision	Avg.
UniCAR-RL(full)	78.5	51.2	80.4	53.3	65.9
w/o Reasoning-RL Branch	77.2	49.8	78.1	51.4	64.2
w/o Caption-RL Branch (Vanilla GRPO)	76.3	48.5	76.0	50.8	62.8
w/o QA-RL Branch (Base Qwen3-VL-4B)	75.0	47.2	73.5	50.1	61.5

Table 5. Branch ablations of UniCAR-RL. By systematically removing each branch, UniCAR-RL progressively degrades to the standard baseline. The default setup uses Qwen3-VL-4B with the Gemini 3 Pro verifier.

Method	Rollout	Total
Vanilla GRPO (QA-RL)	72.3	297.8
Reasoning-RL	68.4	292.7
Caption-RL	305.1	525.5
UniCAR-RL iteration (sum)	445.8	1116.0 ($\sim 3.75\times$)

Table 6. Training Cost Analysis of UniCAR-RL.

each training branch. As shown in Table 6, QA-RL branch and Reasoning-RL branch have training costs comparable to vanilla GRPO, while the additional overhead mainly comes from Caption-RL due to its extra verifier-based evaluation. A UniCAR-RL iteration takes approximately $3.75\times$ the time of a vanilla GRPO step. However, UniCAR-RL requires fewer optimization iterations in the observed training runs, with

about 70 iterations compared with about 150 steps for vanilla GRPO, resulting in an overall training-time ratio of approximately $1.75\times$. Overall, UniCAR-RL achieves substantial performance gains with a moderate increase in training cost.

5. Conclusion

In this paper, we propose UniCAR-RL, a novel RL framework for annotation-free perception optimization that addresses the visual perception bottlenecks of MLLMs in complex visual mathematical reasoning. The core innovation lies in decoupling the optimization of perception and reasoning without relying on data distillation. By introducing a verifier to provide pure visual perception optimization signals and leveraging gold-description-based reasoning to deliver pure logical reasoning signals, UniCAR-RL can precisely co-optimize the model’s fine-grained perception and logical

reasoning capabilities. Extensive experiments show that our method exhibits robust generalization across diverse model architectures and sizes. Notably, because UniCAR-RL’s performance gains stem from breaking the inherent perceptual bottlenecks of MLLMs, particularly in small sizes and earlier architectures, these enhancements naturally generalize to broad multimodal tasks.

Limitations

Within the UniCAR-RL framework, the joint optimization of the synergistic Caption-RL branch, Reasoning-RL branch, and QA-RL branch inevitably increases the training cost, as discussed in Section 4.7. Moreover, the framework is primarily validated on complex mathematical and a limited set of general visual tasks. Future work will explore its application to a broader range of multimodal scenarios.

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A. Visualization of the UniCAR-RL Training Workflow

To provide an intuitive view of UniCAR-RL, Fig. 4 visualizes a training step across the three branches on a geometry problem. The **Caption-RL Branch** generates G image descriptions and assigns per-rollout perception rewards via the advanced verifier, promoting the highest-scoring one as the gold caption. Since the verifier solves purely from text, its accuracy directly reflects faithfulness of visual evidence in the description. The **Reasoning-RL Branch** then reasons over this gold caption with the image withheld, while the **QA-RL Branch** performs standard end-to-end inference from the image and question. The three branches are executed sequentially within each iteration.

B Instruction Templates for UniCAR-RL This section details the prompt templates used in UniCAR-RL training. We design three task-specific templates for the three synergistic branches of our framework, as illustrated in Fig. 5, Fig. 6, and Fig. 7. They correspond respectively to the **QA-RL Branch** for end-to-end answer alignment, the **Caption-RL Branch** for pure visual description extraction, and the **Reasoning-RL Branch** for text-only logical deduction.

QA-RL Branch System Prompt

You FIRST think about the reasoning process as an internal monologue and then provide the final answer. The reasoning process MUST BE enclosed within `<think>` `</think>` tags. The final answer MUST BE put in `\boxed{}`.

Figure 5: System prompt used by the QA-RL Branch.

Caption-RL Branch System Prompt

As a helpful assistant, your role is to support visually impaired users by providing detailed and precise descriptions of input images. Your descriptions should be thorough, enabling users to understand and address any issues related to the images without needing to see them. Focus on offering visual aid without solving the questions for them.

Figure 6: System prompt used by the Caption-RL Branch.

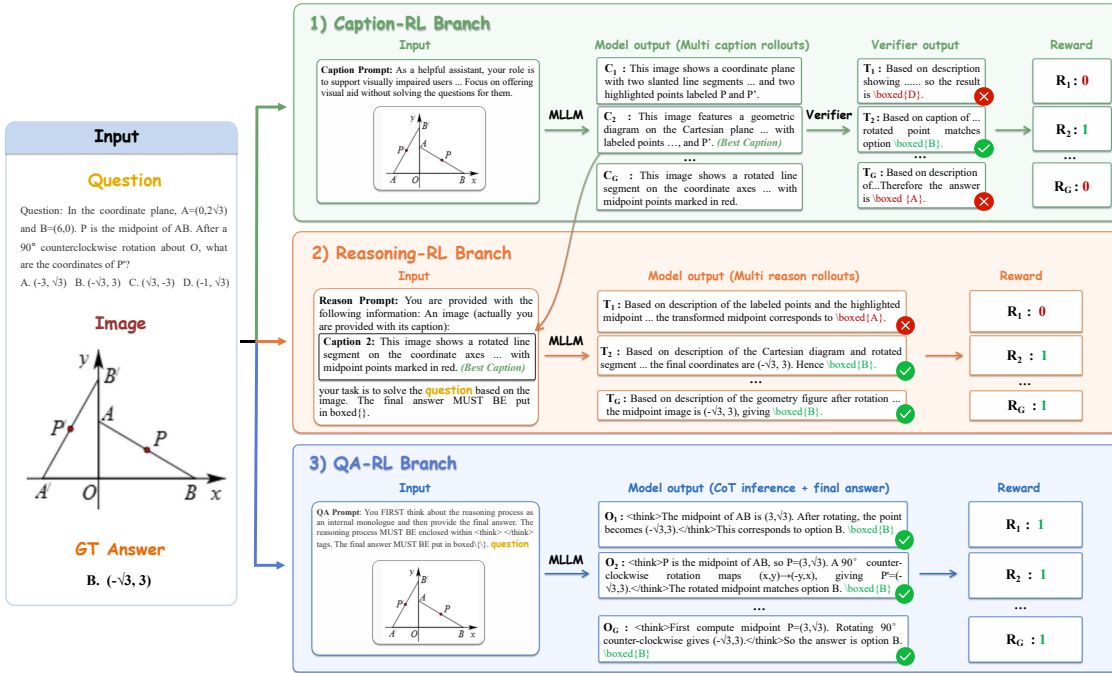


Figure 4: Visualization of a single UniCAR-RL training step. The three branches share the same policy network and are jointly optimized via a unified GRPO objective.

C. Related Work

To complement Section 2, Table 7 compares multimodal mathematical reasoning methods by **training paradigm** (Zero-RL vs. SFT+RL), **distillation** (whether teacher-generated CoT supervision is used, which may introduce additional cost and teacher hallucinations), and **perception** (whether visual perception is explicitly optimized). Existing methods in *RL for Mathematical Reasoning* and *Visual Perception Enhancement* typically focus on only part of these aspects. In contrast, UniCAR-RL combines Zero-RL, avoids distilled supervision, and explicitly optimizes visual perception within a unified framework.

Reasoning-RL Branch System Prompt

You are provided with the following information: an image (actually you are provided with its caption). Your task is to solve the question based on the image. The final answer MUST BE put in \boxed{ }.

Figure 7: System prompt used by the **Reasoning-RL Branch**, where the image is withheld and model reasons purely from caption.

Table 7. Comparison of representative methods for multimodal mathematical reasoning. “Train” indicates the training paradigm, where “Zero-RL” denotes direct RL optimization from a pretrained or instruction-tuned model without task-specific cold-start SFT, “SFT+RL” denotes a cold-start supervised stage followed by reinforcement learning, and “–” indicates non-RL methods. “Distill.” indicates whether the method explicitly relies on distilled reasoning traces, teacher-generated supervision, or synthetic high-quality CoT/QA data. “Percep.” denotes whether the method explicitly enhances visual perception, grounding, diagram understanding, or visual reasoning capabilities.

Method	Venue	Backbone	Size	Train	Distill.	Percep.
<i>Reinforcement Learning for Mathematical Reasoning</i>						
SRPO [19]	NeurIPS’25	Qwen2.5-VL	7B/32B	SFT+RL	✓	✓
Vision-R1 [21]	ICLR’26	Qwen2.5-VL	7B/32B/72B	SFT+RL	✓	✓
MM-Eureka [23]	TMLR’26	Qwen2.5-VL	7B/32B	Zero-RL	×	✓
R1-VL [24]	ICCV’25	Qwen2-VL	7B	SFT+RL	×	✓
URSA [22]	NeurIPS’25	Qwen2.5-Math	8B	SFT+RL	✓	✓
We-Math 2.0 [20]	ICLR’26	Qwen2.5-VL	7B	SFT+RL	✓	✓
VL-Cogito [25]	arXiv’25	Qwen2.5-VL	7B	Zero-RL	×	✓
Shuffle-R1 [4]	ICLR’26	Qwen2.5-VL	3B/7B	Zero-RL	×	✓
<i>Visual Perception Enhancement in MLLMs</i>						
CodePercept [15]	CVPR’26	Qwen3-VL	4B/8B/32B	SFT+RL	×	✓
GeoCode [27]	ICML’26	Qwen2.5/3-VL	7B	SFT	×	✓
MathFlow [26]	ACL’26	Qwen2-VL	7B	SFT	✓	✓
ViRC [28]	CVPR’26	Qwen2.5-VL	7B	SFT+RL	×	✓
Geoint-R1 [29]	CVPR’26	Qwen2.5-VL	7B	SFT+RL	×	✓
Perception-R1 [30]	ICLR’26	Qwen2/2.5-VL	7B	Zero-RL	×	✓
CogFlow [31]	ICLR’26	Qwen2.5-VL	7B	SFT+RL	×	✓
CapRL [32]	ICLR’26	Qwen2.5/3-VL	2B–8B	Zero-RL	×	✓
VPPO [33]	ICLR’26	Qwen2.5-VL	7B/32B	Zero-RL	×	✓
Vision-SR1 [14]	ICLR’26	Qwen2.5-VL	7B	Zero-RL	×	✓
PAP0 [34]	ICLR’26	Qwen2.5/3-VL	2B/3B/7B	Zero-RL	×	✓
NoisyRollout [35]	NeurIPS’25	Qwen2.5-VL	7B	Zero-RL	×	✓
UniCAR-RL (Ours)	–	Qwen2.5-VL, Qwen3-VL	3B/4B/7B/8B	Zero-RL	×	✓

Table 8. Additional perception results on ChartQA, HallBench, V*Bench and HR-Bench. Gemini 3 Pro is used as the verifier during training by default.

Model	ChartQA	HallBench	V*Bench	HR-Bench 4K	HR-Bench 8K
Qwen3-VL-4B	83.0	73.7	86.4	80.0	73.6
+UniCAR-RL	89.5	75.5	88.5	81.6	76.4
Qwen3-VL-8B	88.4	75.0	88.7	81.2	76.5
+UniCAR-RL	90.6	75.4	90.1	82.8	78.3

D. Additional Results on General Benchmarks

To further evaluate the generalization of UniCAR-RL beyond visual mathematical reasoning, we extend the evaluation to a broader range of general visual tasks and model configurations. Beyond ChartQA and HallusionBench reported in Section 4.4, we further evaluate on V*Bench [49] and HR-Bench [50], covering complementary aspects of general visual understanding, including fine-grained visual search and high-resolution perception. As shown in Table 8, UniCAR-RL consistently improves performance across these additional tasks and architectures, further demonstrating its ability to generalize from visual mathematical reasoning to a broader range of general visual tasks.

E. Prompts for General Benchmarks

For reproducibility, we provide the prompts used for evaluating the general multimodal benchmarks. All benchmarks are evaluated under the same chat template, with the following shared instruction prepended to the benchmark-specific question:

You FIRST think about the reasoning process as an internal monologue and then provide the final answer. The reasoning process MUST BE enclosed within <think> </think> tags. The final answer MUST BE put in \boxed{ }.

The image is then provided together with the benchmark-specific question. The corresponding question templates are described in Fig. 8.

ChartQA We directly use the original question without any task-specific modification. {question}	HallusionBench We retain the original question and explicitly constrain the final response to a binary answer. {question} Your final answer can only be yes or no.
V*Bench Each question is formatted as a four-way multiple-choice problem. Question: {question} Options: A. {option A} B. {option B} C. {option C} D. {option D} Please select the correct answer from the options above.	HR-Bench HR-Bench-4K and HR-Bench-8K use the same four-way multiple-choice prompt format. Question: {question} Options: A. {option A} B. {option B} C. {option C} D. {option D} Please select the correct answer from the options above.

Figure 8. **Task-specific prompt templates and output evaluation protocols.** Placeholders enclosed in braces are replaced with the corresponding benchmark questions and answer options at inference time.

For output evaluation, ChartQA answers are extracted and evaluated using the benchmark-specific answer matching procedure. For HallusionBench, we extract the answer from the final boxed response and match it to `yes/no`. For V*Bench and HR-Bench, the predicted option letter is extracted and compared with the ground-truth choice.